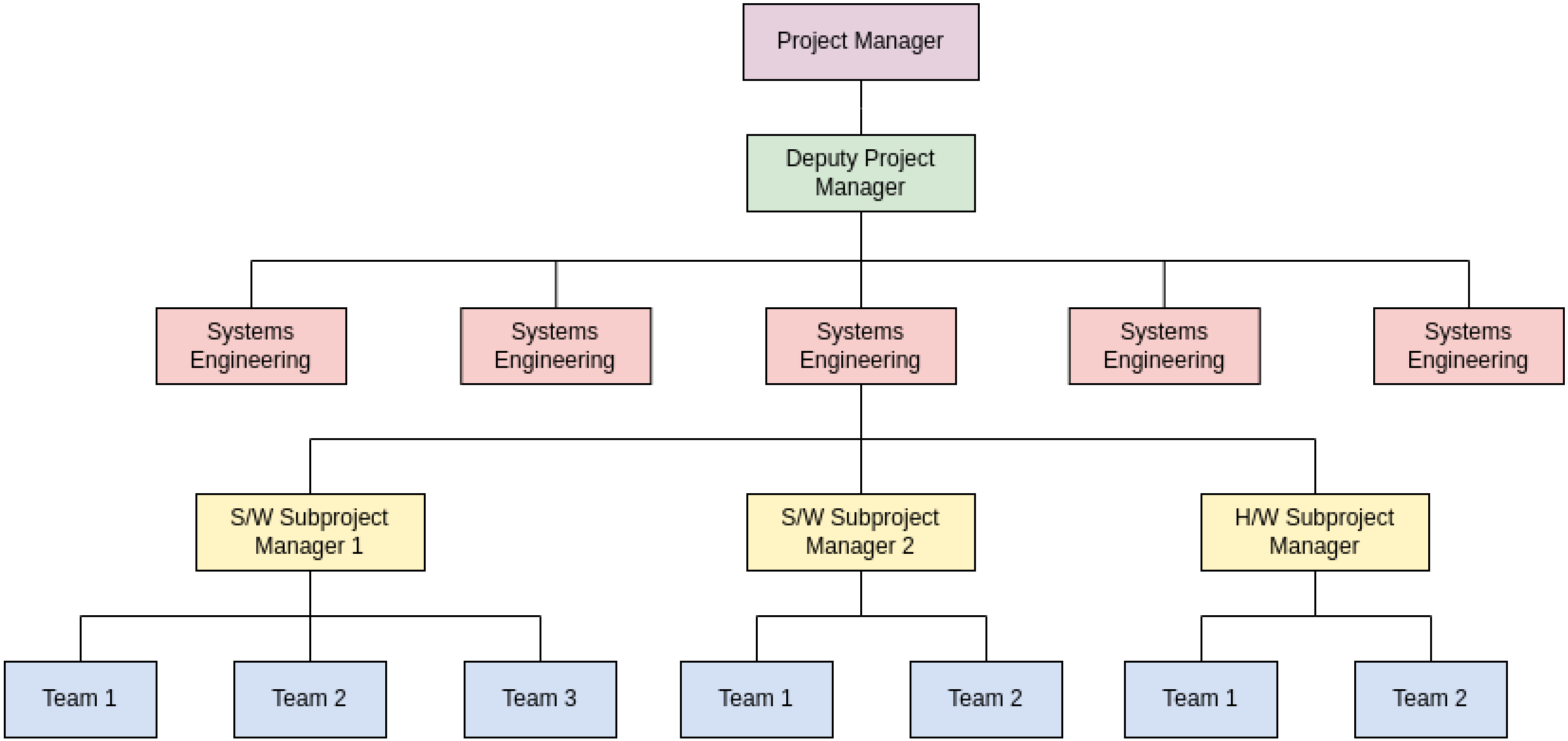
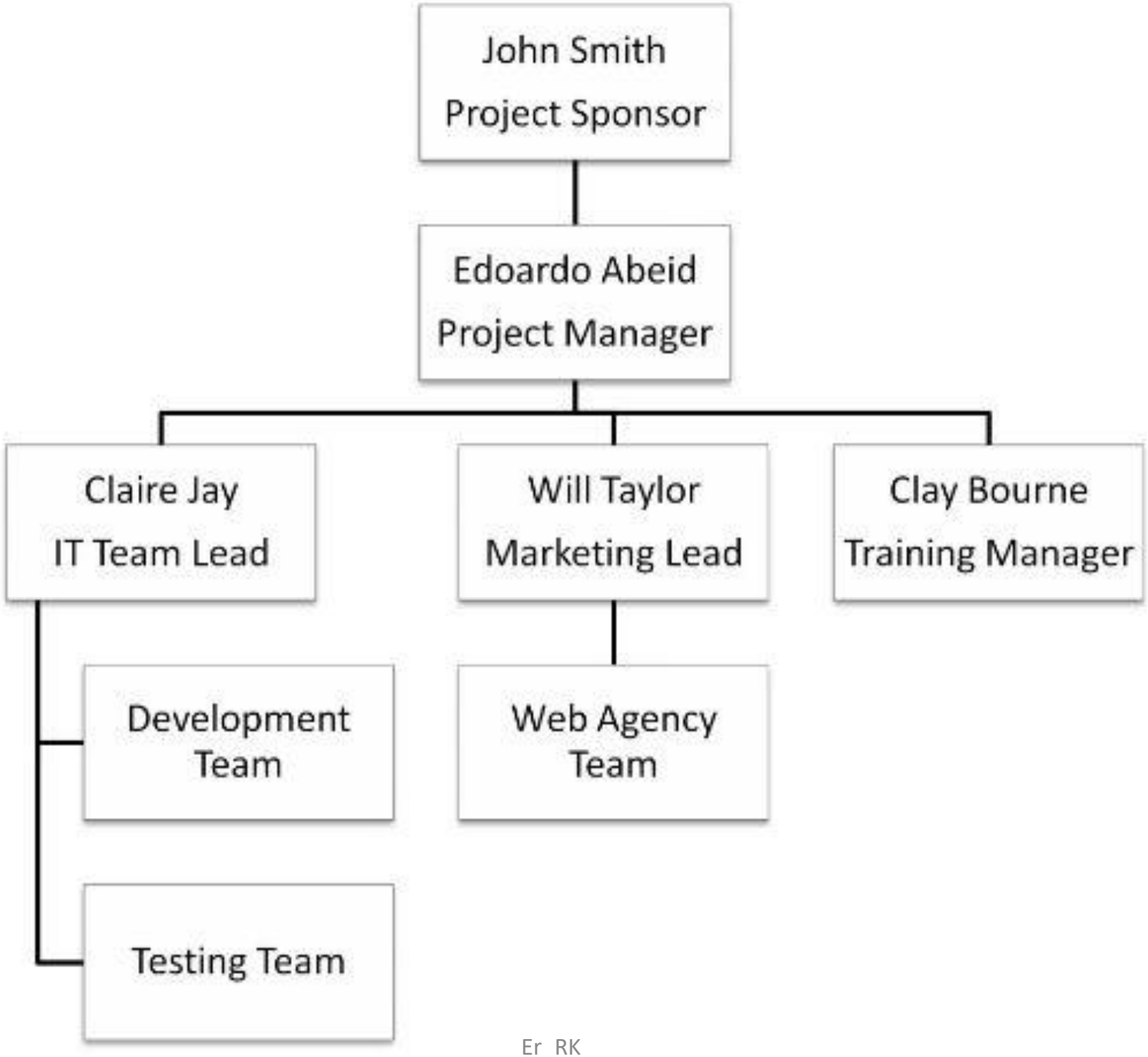
Project Organization and Responsibilities

Unit_4


Project Manager
Deputy Project Manager
Systems Engineering
Systems Engineering
Systems Engineering
Systems Engineering
Systems Engineering
S/W Subproject Manager 1
S/W Subproject Manager 2
H/W Subproject Manager
Team 1
Team 2
Team 3
Team 1
Team 2
Team 1
Team 2


John Smith
Project Sponsor
Edoardo Abeid
Project Manager
Claire Jay
IT Team Lead
Will Taylor
Marketing Lead
Clay Bourne
Training Manager
Development Team
Web Agency Team
Testing Team

Meaning of Project Organization

- Project organization defines how people, roles, and reporting lines are arranged so that a project can be planned, executed, and completed in a controlled manner.
- A well-designed structure prevents overlap, avoids delays, and assigns clear accountability.
- A project is temporary, so its organization focuses on efficiency, clarity, and coordination rather than long-term hierarchy.

Meaning of Project Organization

- Project organization refers to the way people, roles, and responsibilities are arranged to complete a project.
- It helps define **who does what**, **how decisions are made**, and **how information flows** within a project.
- **Why Project Organization Is Needed?**
  - Ensures clear responsibility
  - Reduces confusion during work
  - Improves coordination between teams
  - Helps monitor progress and solve problems quickly
  - Supports timely decision-making

2. Major Roles and Their Responsibilities

- **A. Project Sponsor**
- The sponsor is the senior authority who approves the project and supports it throughout.
- **Key responsibilities**
  - Sanctions budget and resources
  - Approves major changes
  - Clears obstacles that the project manager cannot solve
  - Ensures the project supports overall organizational goals
- **Example for validation**
  - A hydropower project in Nepal (such as a 25 MW project funded by a local investor group) needs approval for tunnel construction. The sponsor ensures the funds are available and approves the engineering changes. Without the sponsor, the project manager cannot move ahead with procurement or major structural adjustments.

- **B. Project Manager**
  - The project manager coordinates the work from start to finish.
- **Key responsibilities**
  - Creates the project plan
  - Assigns tasks to team members
  - Manages cost, time, and scope
  - Conducts risk analysis and mitigation
  - Communicates with stakeholders
  - Ensures the work meets quality expectations
- **Example for validation**
  - In a road construction project under the Department of Roads, the project manager plans the schedule for surveying, design, and construction. If heavy rainfall delays excavation, the manager adjusts the timeline and reallocates resources to maintain progress. This illustrates active control of time and resources.

- **C. Project Team Members**
- These are specialists or workers who perform the actual tasks.
- **Responsibilities**
- Complete assigned activities
- Report progress and issues
- Follow quality and safety standards
- Cooperate with other teams
- **Example**
- During the development of a mobile banking application in Nepal, team programmers build specific modules such as login, fund transfer, and QR payments. One developer identifies a security loophole and reports it early, preventing later delays. This shows the importance of responsibility and early communication.

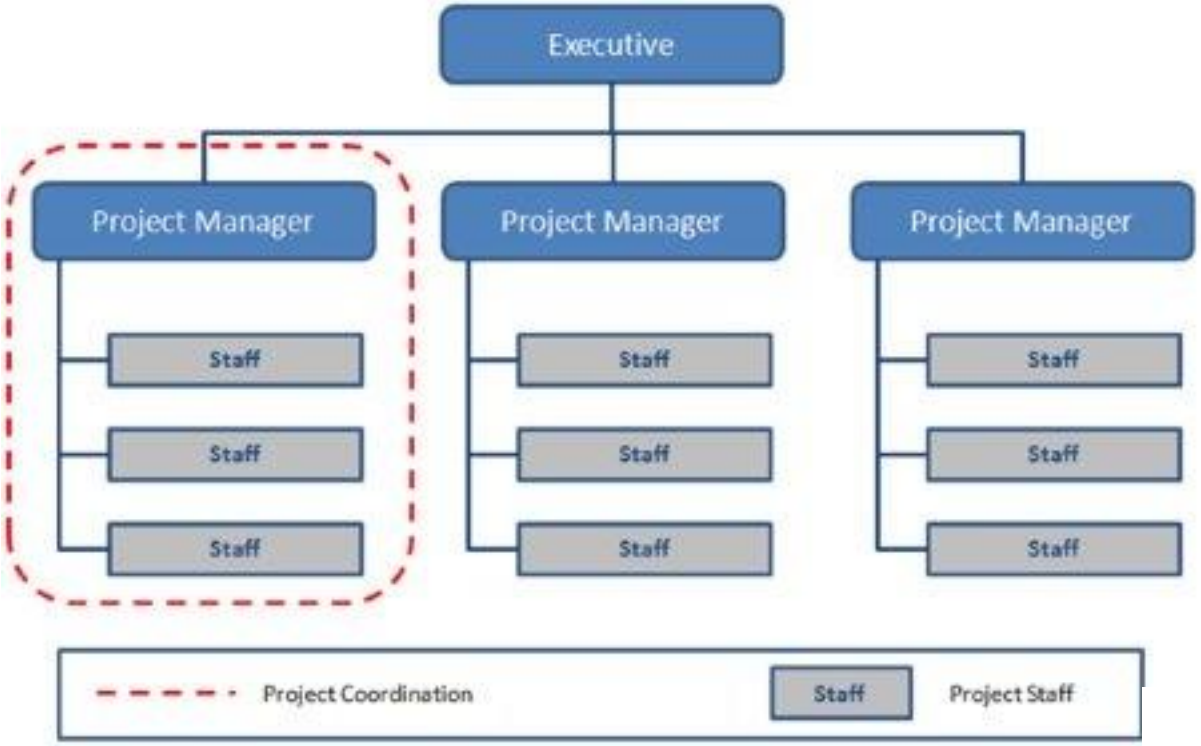
- **D. Stakeholders**
- Stakeholders include individuals, groups, or institutions that affect or are affected by the project.
- **Responsibilities**
- Provide requirements
- Give feedback
- Monitor compliance with regulations
- Approve outputs when necessary
- **Example**
- A hospital expansion project in Pokhara requires approval from the local municipality, the health ministry, and nearby residents affected by traffic. Their role is not technical work but coordination, review, and permission. This confirms that stakeholders help shape and validate project decisions.

- **E. Support Functions**
- These are units that help the project operate smoothly.
- **Common support groups**
- Finance (budget handling)
- HR (recruitment and contracts)
- Procurement (materials and vendors)
- IT support (systems and tools)
- Legal and compliance
- **Example**
- In a software project for a private college in Nepal, the procurement unit handles hardware purchases for servers. This keeps the project team focused on development rather than administrative tasks.

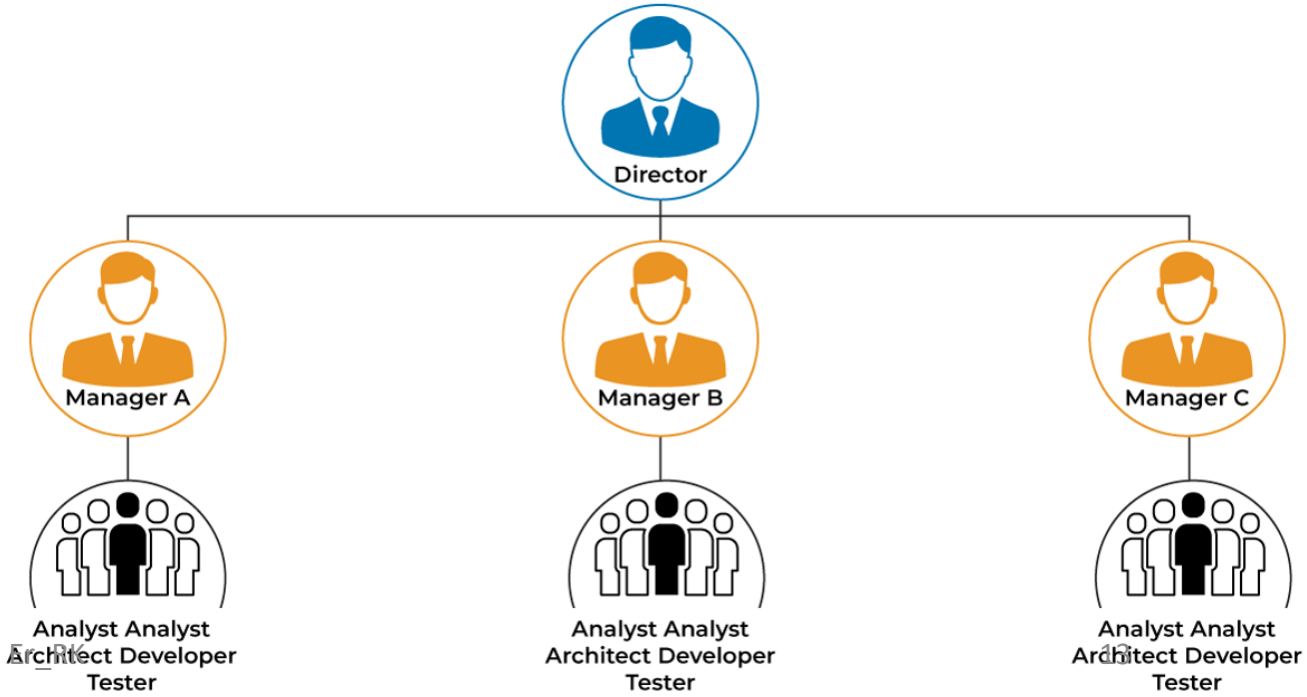
Types of Project Organization Structures

- When designing a project organization inside an existing institution, the most important factor is **how much authority top management is willing to give to the project manager**.
- Top management decides:
  - What the project manager can approve
  - How teams will be formed
  - How responsibilities will be divided
  - How communication will flow
  - How conflicts between project and departments will be resolved

- Development organizations (such as NGOs, government programs, health or education sectors) usually work in **program units**. Within these units, three project organization choices are available:
  1. **Programmatic Based**
  2. **Matrix Based**
  3. **Project Based**
- Each structure is defined by the *level of authority given to the project manager*.



Programmatic Based Organization Structure



Programmatic Based Organization

- A programmatic structure is the traditional arrangement where project resources come from one program sector (such as health, education, agriculture, etc.).
- The project manager has authority only **within that program area**.
- Example:
- A **health-focused NGO** places a maternal nutrition project inside its Health Unit.
- All staff (nutritionists, field workers, trainers) come from the same unit.

Characteristics

- Project teams are staffed only from one program sector
- Team members usually know each other and share similar skills
- Clear reporting lines
- Project manager often doubles as program unit manager

- **Advantages**

1. **Clear authority**
    1. The program manager controls all resources needed for the project.
2. **Easy coordination**
    1. Since staff belong to the same department, communication is simple.
3. **Shared expertise**
    1. Members already understand the subject area, reducing training time.

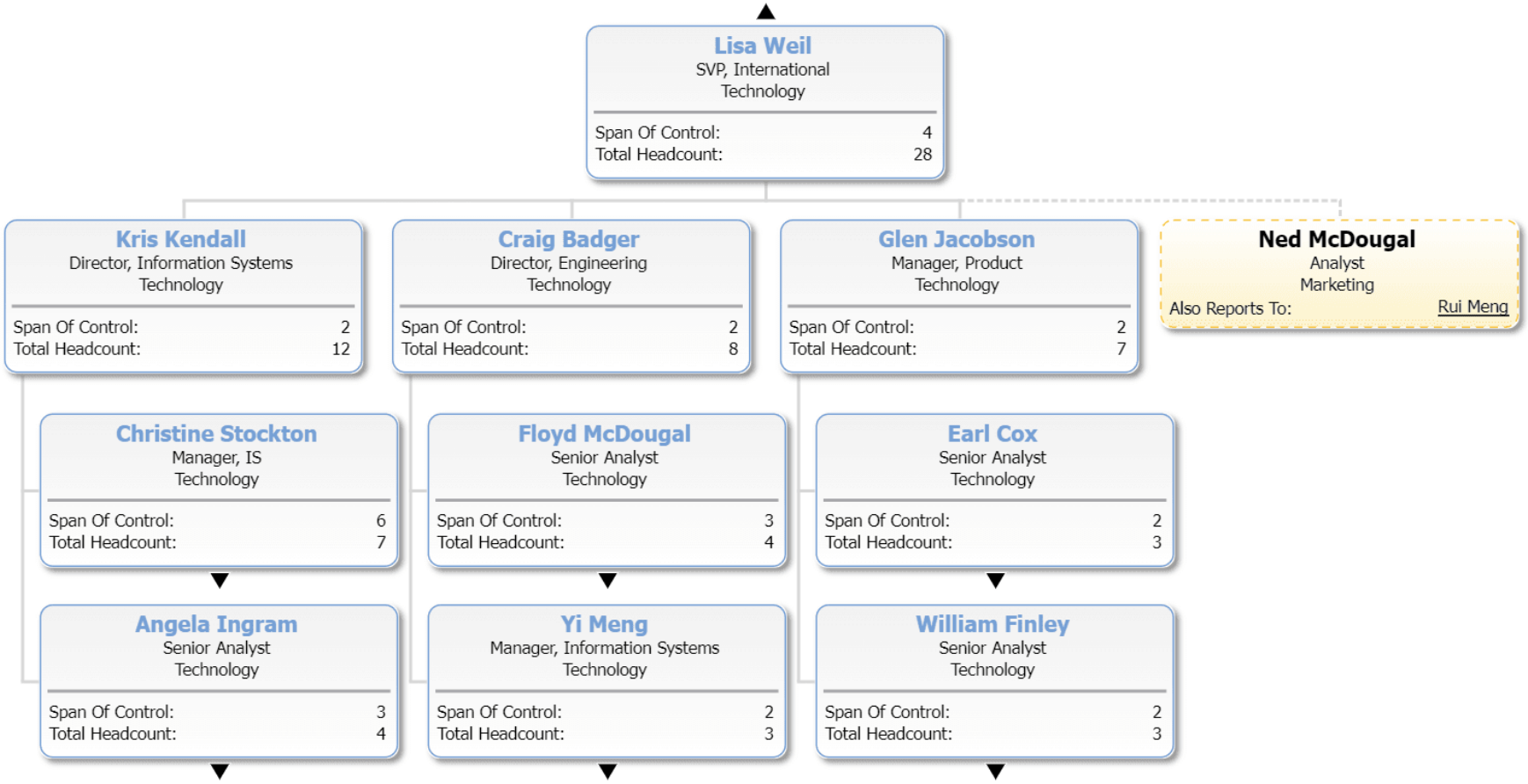
- **Disadvantages**

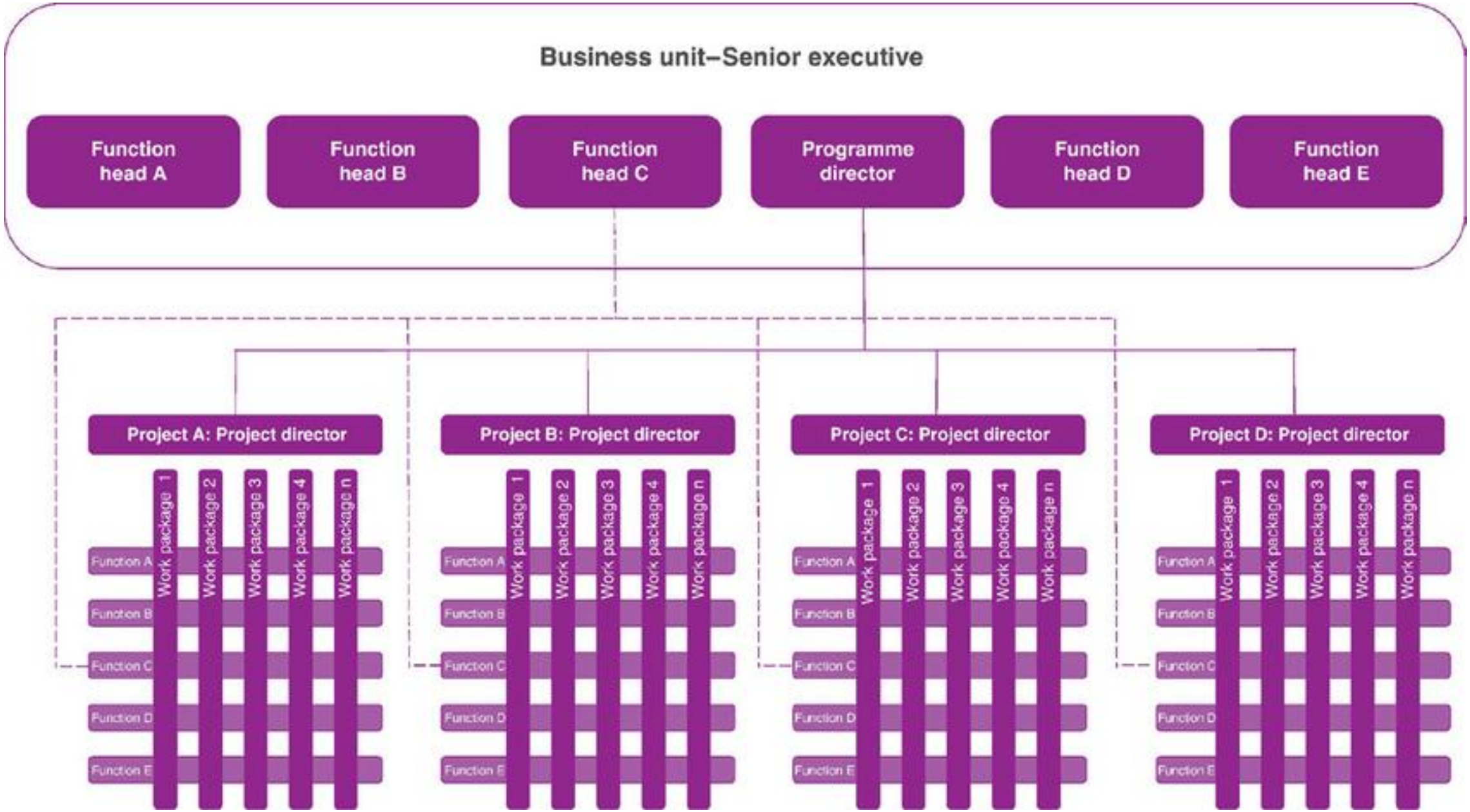
1. **Limited skill diversity**
    1. If the project needs multi-sector expertise, the unit may not have the needed specialists.
2. **Part-time project participation**
    1. Staff often have other program duties, creating scheduling conflicts.
3. **Slow progress on cross-sector tasks**
    1. The unit may struggle when the project involves other disciplines.

- **Example:**
- A **nutrition improvement project** under the Ministry of Health needs both nutritionists and civil engineers (for water system repair). Since the Health Unit does not have civil engineers, the project struggles to find the needed skill. Work gets delayed because authority lies only within the health sector.
- **Validation:**
- This proves that the programmatic structure works only for single-sector projects and becomes weak when cross-sector skill is required.

Matrix Based Organization

- A matrix structure allows staff to work under two directions:
  - Their **functional/program manager**
  - A **project manager** for specific projects
  - Staff may be assigned temporarily to projects depending on expertise.
- Example:
- A monitoring and evaluation (M&E) specialist reports to the M&E Unit but works part-time on four different NGO projects.
- **Characteristics**
  - People report to both functional units and project managers
  - Resources are shared across multiple projects
  - Allows a wide mix of expertise
  - Most flexible structure for development organizations


Lisa Weil
SVP, International Technology
Span Of Control: 4
Total Headcount: 28
Kris Kendall
Director, Information Systems Technology
Span Of Control: 2
Total Headcount: 12
Craig Badger
Director, Engineering Technology
Span Of Control: 2
Total Headcount: 8
Glen Jacobson
Manager, Product Technology
Span Of Control: 2
Total Headcount: 7
Ned McDougal
Analyst Marketing
Also Reports To: Rui Meng
Christine Stockton
Manager, IS Technology
Span Of Control: 6
Total Headcount: 7
Floyd McDougal
Senior Analyst Technology
Span Of Control: 3
Total Headcount: 4
Earl Cox
Senior Analyst Technology
Span Of Control: 2
Total Headcount: 3
Angela Ingram
Senior Analyst Technology
Span Of Control: 3
Total Headcount: 4
Yi Meng
Manager, Information Systems Technology
Span Of Control: 2
Total Headcount: 3
William Finley
Senior Analyst Technology
Span Of Control: 2
Total Headcount: 3


Business unit–Senior executive
Function head A
Function head B
Function head C
Programme director
Function head D
Function head E
Project A: Project director
Project B: Project director
Project C: Project director
Project D: Project director
Work package 1
Work package 2
Work package 3
Work package 4
Work package n
Function A
Function B
Function C
Function D
Function E

- **Advantages**

1. Efficient use of scarce experts

1. Example: One water engineer can support four rural water projects rather than sitting idle in one project.

2. Better knowledge sharing

1. Staff interact across departments, improving collaboration.

3. Supports professional development

1. Specialists maintain ties with their functional units and continue learning.

4. Useful in dynamic environments

1. When donor priorities change, staff can be reallocated quickly.

- **Disadvantages**

1. Complex reporting

1. Staff may receive instructions from two managers, which can cause confusion.

2. Competition for resources

1. Multiple projects may need the same expert at the same time.

3. Higher coordination effort

1. Project managers must negotiate for resources rather than directly controlling them.

4. Requires strong communication and time management

- **Example:**
- A national NGO running projects in **Education**, **Health**, and **WASH** shares specialists across all projects.
- A data analyst works on an education dashboard, a health tracking tool, and a water quality assessment
- Each project manager needs the analyst at the same time
- Coordination meetings are needed to decide priority
- **Validation:**
- This demonstrates that matrix structures make the best use of experts but require careful scheduling and cooperation.

Project Based Organization

- The project manager has **full authority** over all project resources. Team members are assigned **exclusively** to the project and report only to the project manager.
- This structure is common in **large, complex, or donor-funded projects**.
- Example:
- A **large hydropower project** in Nepal (such as Upper Tamakoshi or Arun III) forms a completely separate project unit with its own engineers, HR staff, procurement staff, and finance officers.

Projectized Organization Structure

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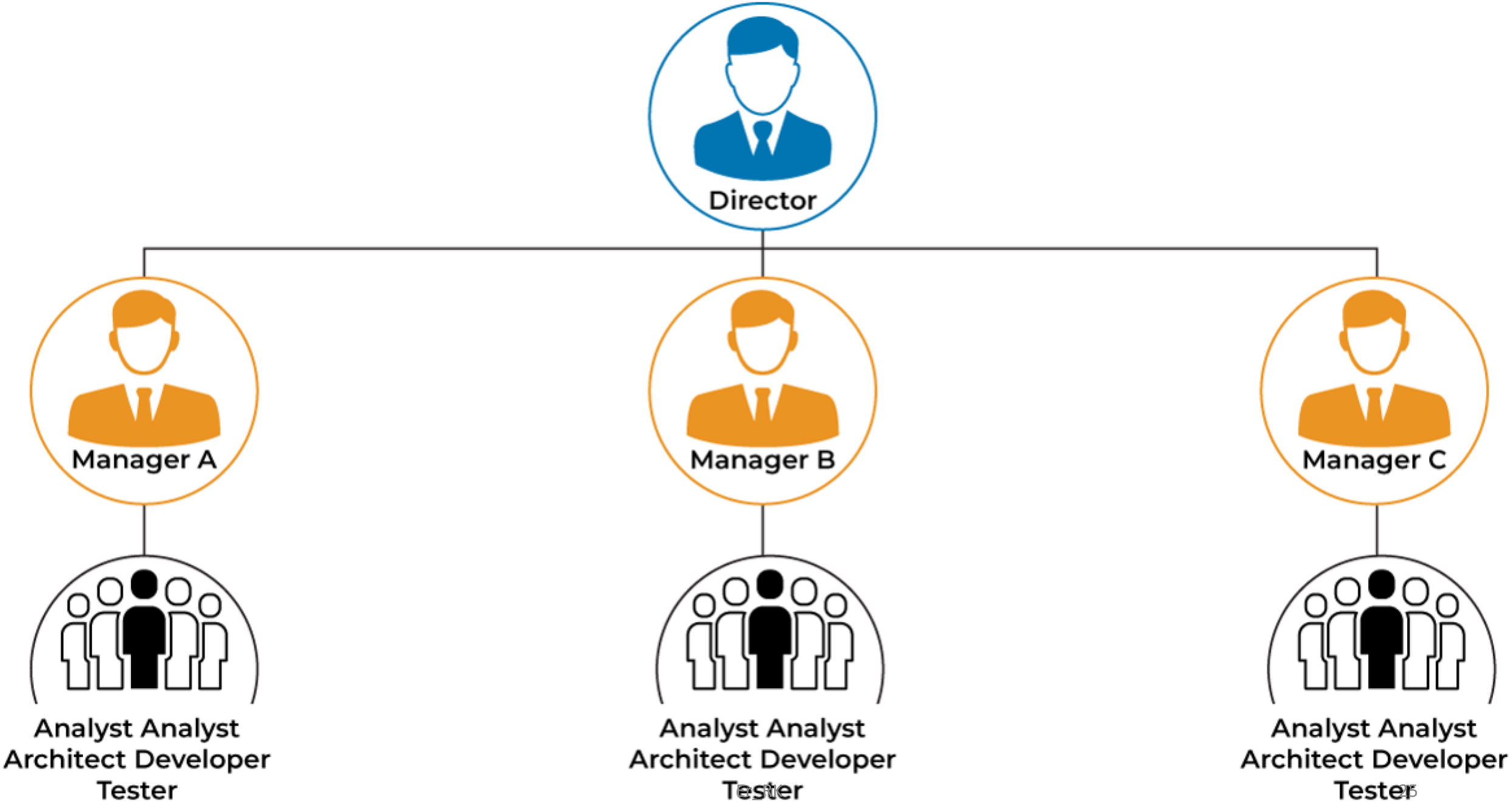

Director
Manager A
Manager B
Manager C
Analyst Analyst
Architect Developer
Tester
Analyst Analyst
Architect Developer
Tester
Analyst Analyst
Architect Developer
Tester

Characteristics

- Project manager has total control over planning, staffing, and decision-making
- Team members have strong project loyalty
- Communication is centralized
- Fast response to issues
- Little dependence on functional departments

- **Advantages**

1. Fast decision-making

1. The project manager approves resources immediately.

2. High team commitment

1. Staff work full-time on the project and understand goals deeply.

3. Clear accountability

1. No confusion about authority or responsibility.

4. Strong control over cost, schedule, and quality

- **Disadvantages**

1. High cost of staffing

1. Full-time teams mean more resources than needed during slow periods.

2. Duplication of roles

1. Example: Every large project may need its own finance officer or engineer, increasing overall cost.

3. Knowledge isolation

1. Lessons learned may not flow easily to other projects.

4. Uncertain career progression

1. When a project ends, staff may not know where they will be reassigned.

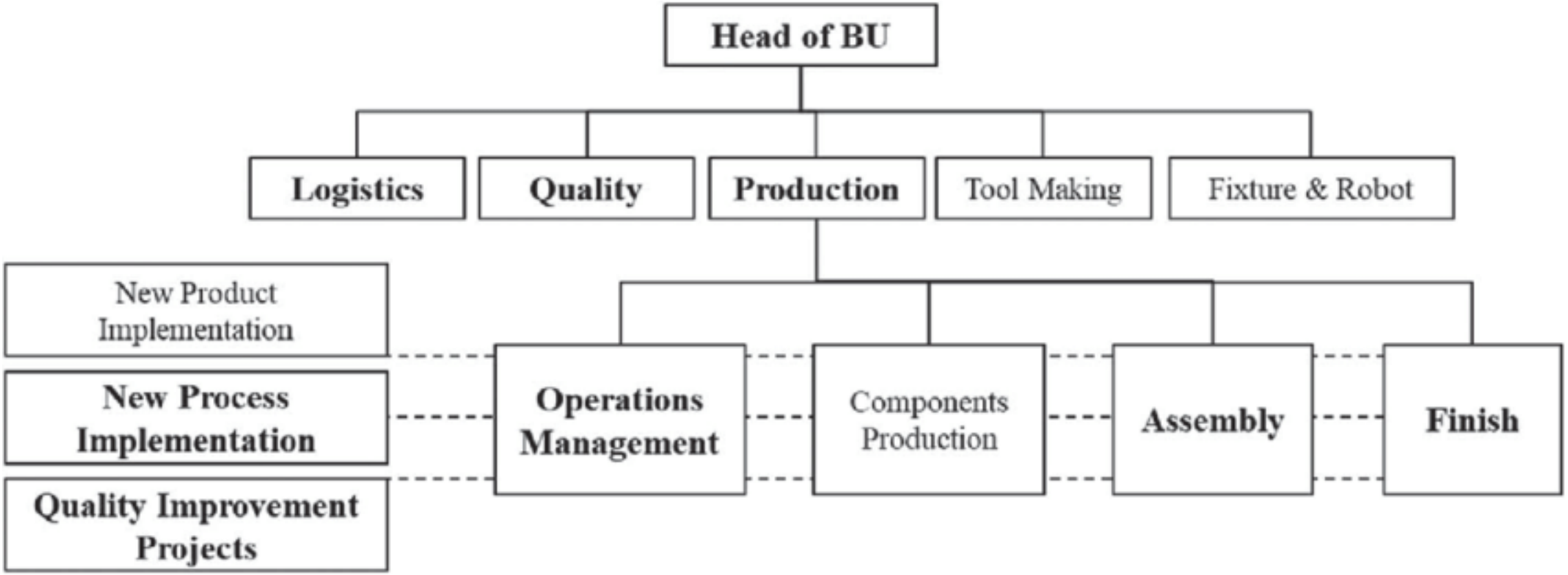
- **Example:**
- A donor-funded **Water Supply and Sanitation Project** forms a project-based unit.
- The project hires engineers, social mobilizers, accountants, and procurement specialists
- They work only on this project for 5 years
- When the project ends, contracts finish and staff must find new assignments
- **Validation:**
- This confirms both strengths (clear authority, fast decisions) and weaknesses (resource duplication, end-of-project job uncertainty).

Why Clear Responsibilities Matter

- **A. Reduces duplication of work**
  - Example: Two survey teams preparing the same map for a road project wastes money. A clear responsibility chart avoids this.
- **B. Improves accountability**
  - Example: If a delay occurs in a hydropower transmission line, the responsibility matrix helps identify whether the issue belongs to design, procurement, or civil works.
- **C. Supports quality control**
  - Example: In a software project for a Nepali college, quality assurance is assigned to a dedicated tester. This ensures independent checking rather than relying on programmers.
- **D. Helps manage risk**
  - Example: During monsoon, a construction project assigns a risk officer to monitor landslide-prone areas, preventing structural damage.

Line of Business (LOB) Organization

- The **Line of Business (LOB)** structure groups activities based on major product or service lines. Each line operates almost like a small company inside the larger organization.
- **Key Characteristics**
- Each line manages its own operations, budget, and strategic goals
- Decision-making is faster because authority stays within the unit
- LOB leaders are responsible for performance
- **Example**
- Telecom companies like **NTC** have separate LOB units such as GSM service, FTTH service, and Customer Care.
- Banks divide operations into Retail Banking, SME Banking, and Digital Banking.
- Each line has its own head who is accountable for growth, customer satisfaction, and operational improvement.


Head of BU
Logistics
Quality
Production
Tool Making
Fixture & Robot
New Product Implementation
New Process Implementation
Quality Improvement Projects
Operations Management
Components Production
Assembly
Finish

Core Idea of LOB Structure

- In a traditional vertical structure, the organization is arranged by functions such as HR, finance, or operations. In an LOB structure, the arrangement follows the **type of service or product** provided.
- Each line is responsible for:
  - Its own strategic direction
  - Its own financial performance
  - Development of specialized services
  - Managing customer experience
  - Coordinating projects that belong to that business line

Characteristics

- **A. Independent business units**
  - Each LOB operates almost like a mini-organization with its own manager, budget, staff, and performance measures.
- **B. Strong customer orientation**
  - LOBs focus on meeting the needs of a specific customer group or market.
- **C. Focused decision-making**
  - Managers of each line make quick decisions because authority is concentrated within that business line.
- **D. Alignment of resources**
  - Human resources, technology, and investments are grouped according to the line's goals.
- **E. Performance measured separately**
  - Each business line tracks its revenue, cost, and growth independently.

How LOB Works in Daily Operations

- A company may have several business lines. For example:
- Retail Banking
- SME Banking
- Digital Banking
- Remittance Services
- Each line maintains its own projects, staff, and operational priorities. They report performance to top management but operate with considerable autonomy.

Roles and Responsibilities Within an LOB

- **A. LOB Manager**
- Leads the entire line of business
- Sets targets and strategies
- Controls budget
- Approves major projects within the line
- Coordinates with top management
- **B. Project Managers under LOB**
- Handle projects linked to the business line
- Report to the LOB Manager
- Manage staff, vendors, deadlines, and outputs

C. Operational Staff

•Provide daily service delivery based on the line's specialty

•Engage directly with customers

•Maintain work quality standards

D. Support Units (Shared Across Organization)

•HR, finance, IT, and legal support the LOB as needed

•Provide resources, systems, and compliance guidance

Line of Business (LOB) Organization

- **LOB Manager:** Oversees operations, revenue, and performance
- **Project Manager (within LOB):** Leads project-based work tied to that business line
- **Operational Teams:** Handle day-to-day service delivery
- **Support Teams:** Provide HR, IT, procurement help

- **Advantages**
- Clear accountability
- Better focus on customer needs
- Faster decisions
- Encourages innovation within each business line

- **Challenges**
- Risk of duplication of resources
- Internal competition
- Need strong coordination

A. Telecom Sector Example: NTC or Ncell

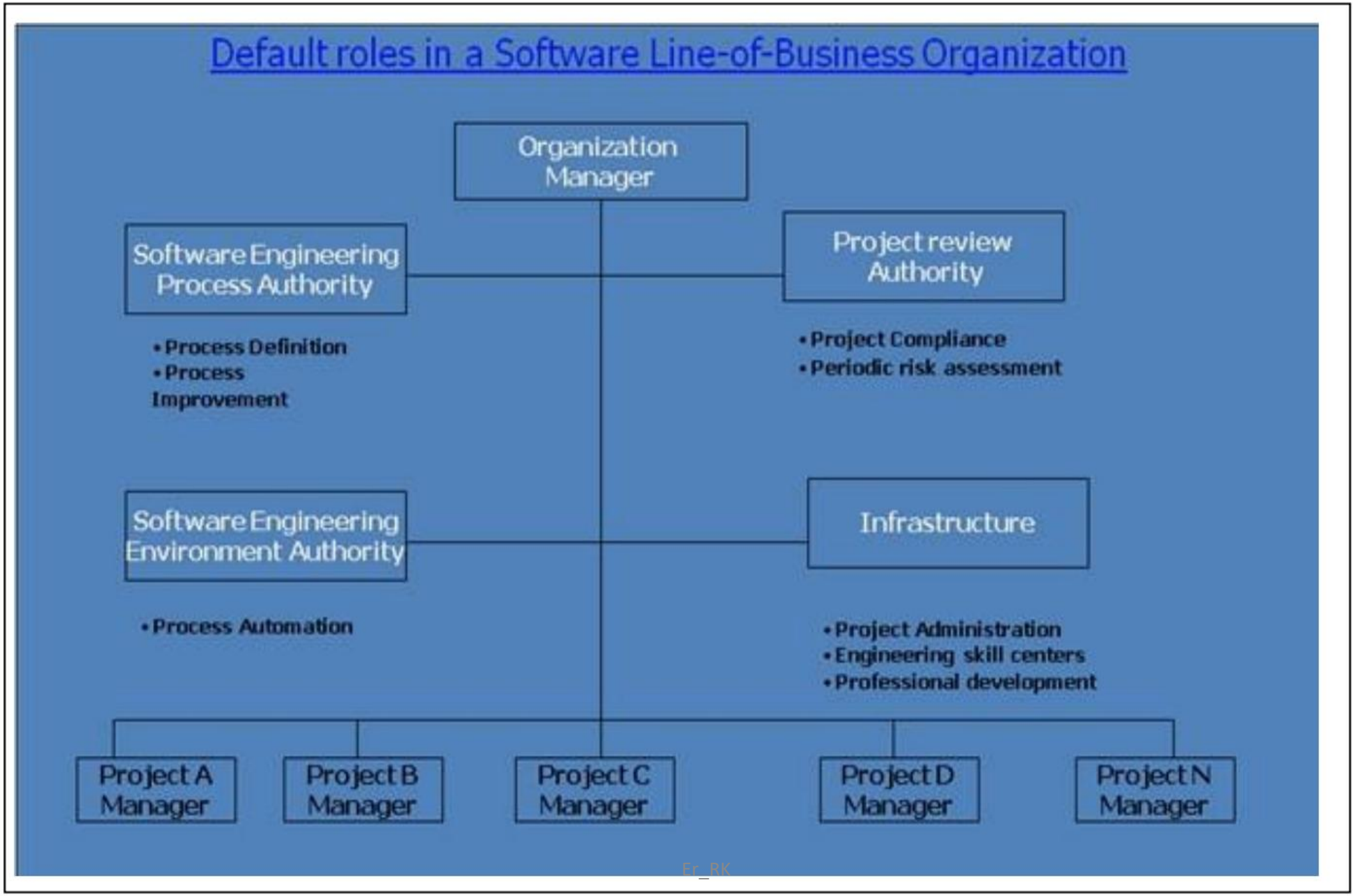
- They have multiple business lines:
  - GSM Mobile Service
  - Internet and FTTH Services
  - Enterprise Solutions
  - Prepaid and Postpaid Billing
- Each business line has:
  - Dedicated manager
  - Separate technical and marketing teams
  - Its own customer support processes
- **Validation:**
- If the GSM unit wants to introduce VoLTE, it runs its own project and budget plan without affecting FTTH or enterprise services.

LOB in Project Management Context

- When a project arises within a business line:
  - The **LOB manager** becomes the project sponsor
  - The project aligns with the LOB's strategic goals
  - A project manager is appointed for execution
  - Other LOB staff provide subject expertise
  - Support units (finance, IT, HR) provide shared services
- **Example**
- A Digital Banking LOB launches an online loan-application platform. The LOB manager approves the project, assigns internal IT staff, and coordinates with compliance and risk units. This shows how LOB integrates strategy with project work.

Main features of default organization

- Responsibility for process definition & maintenance is specific to a cohesive line of business.
- Responsibility for process automation is an organizational role & is equal in importance to the process definition role.
- Organizational role may be fulfilled by a single individual or several different teams.


Default roles in a Software Line-of-Business Organization
Organization Manager
Software Engineering Process Authority
• Process Definition
• Process Improvement
Project review Authority
• Project Compliance
• Periodic risk assessment
Software Engineering Environment Authority
• Process Automation
Infrastructure
• Project Administration
• Engineering skill centers
• Professional development
Project A Manager
Project B Manager
Project C Manager
Project D Manager
Project N Manager

- **1. Organization Manager**
  - This is the person who oversees the entire software business unit.
- **Simple Meaning**
  - This manager makes sure all projects, processes, tools, and teams work together smoothly.
- **Example**
  - In a Nepali software company building online banking systems, the **Head of Software Division** plays this role.
  - They ensure:
  - All projects follow the same standards
  - Teams get required tools and resources
  - Risks are monitored

- **2. Software Engineering Process Authority**
  - This group defines and improves how software is developed across all projects.
- **Simple Meaning**
  - They decide **how** the software should be built.

    They create standard procedures so that all teams follow the same coding and testing practices.
- **Key Tasks**
  - Define development processes
  - Improve workflows over time
- **Example**
- If the company decides that every project must use **Agile development**, this authority creates the rules:
  - How sprints should work
  - How code reviews should be done
  - How testing should be documented
  - This ensures Project A, B, C, and D follow the same process.

- **3. Project Review Authority**
  - This group regularly checks whether each project is following rules and progressing correctly.
- **Simple Meaning**
  - They act like an internal inspection team. They make sure projects are compliant and identify risks early.
- **Key Tasks**
  - Verify project progress
  - Check compliance with standards
  - Conduct periodic risk assessments
- **Example**
- A banking app project must follow strict security rules.
  If Project C misses encryption steps, this authority alerts the organization manager so corrections can be made early.

- **4. Software Engineering Environment Authority**
  - This group manages the tools and automation systems used by all project teams.
- **Simple Meaning**
  - They maintain the software tools needed for development, testing, and deployment.
- **Key Tasks**
  - Process automation
- Managing development tools (e.g., Git, CI/CD systems, testing frameworks)
- **Example**
- They install and support:
  - GitLab or GitHub
  - Jenkins for automation
  - Jira for task management
  - Testing tools
  - When Project A needs a new testing server, this team sets it up.

- **5. Infrastructure Authority**
  - This group handles supporting activities like:
  - Administrative work
  - Skill development
  - Training
  - Internal engineering resources
- **Simple Meaning**
  - They make sure the teams have the right people, skills, and administrative support.
- **Key Tasks**
  - Training employees
  - Managing engineering skill centers
  - Maintaining shared infrastructure
- **Example**
- If the company wants developers to learn cloud computing:
- This authority organizes AWS or Azure training
- It ensures all projects have cloud access and servers

- **6. Project Managers (Project A, B, C, D, N, etc.)**
  - These managers lead individual software projects.
- **Simple Meaning**
  - Each project manager handles one software product or client assignment.
- **Their Work Includes**
  - Planning work
  - Managing teams
  - Coordinating with all authorities mentioned above
- **Example**
- Project A Manager.
  - Handles the development of a **mobile wallet app**.
- Project B Manager:
  - Works on a **college management system** for Nepal.
- Project D Manager:
  - Leads a **government e-procurement portal**.
- All these managers receive support from the Process Authority, Review Authority, Environment Authority, and Infrastructure teams.

- **Software Engineering Process Authority (SEPA):**
  - The SEPA facilities the exchange of information & process guidance both to & from project practitioners.
  - This role is accountable to General Manager for maintaining a current assessment of the organization's process maturity & its plan for future improvement.
- **Project Review Authority (PRA**):
  - The PRA is the single individual responsible for ensuring that a software project complies with all organizational & business unit software policies, practices & standards A software Project
  - Manager is responsible for meeting the requirements of a contract or some other project compliance standard.
- **Software Engineering Environment Authority(SEEA**):
  - The SEEA is responsible for automating the organization's process, maintaining the organization's standard environment, Training projects to use the environment & maintaining organization-wide reusable assets The SEEA role is necessary to achieve a significant ROI for common process.
- **Infrastructure:**
  - An organization's infrastructure provides human resources support, project-independent research & development, & other capital software engineering assets.